

The Pragmatic Synthesis

How Complex Systems Navigate Incompatible Truths
through Adaptive Heuristics

*Why the Structure Called “Consciousness” Is a Necessary
Artefact of Navigation, Not a Separate Ontological Category*

Boris Kriger^{1,2}

ORCID: [0009-0001-0034-2903](https://orcid.org/0009-0001-0034-2903)

¹ Information Physics Institute, Gosport, Hampshire, United Kingdom
boris.kriger@informationphysicsinstitute.net

² Institute of Integrative and Interdisciplinary Research, Toronto, Canada
boriskruger@interdisciplinary-institute.org

Abstract

The first two articles of this monograph established that any sufficiently complex system necessarily harbours content inaccessible to internal observers (Article 1) and admits mutually exclusive but individually valid descriptions that cannot be unified (Article 2). The present article asks the constructive question: how does an agent embedded in such a system manage to function?

We develop a formal theory of *adaptive navigation* among incompatible descriptions. Five primitive notions—agent, working model, switching, cost, and viability—are introduced together with four axioms (Computational Boundedness, Context Multiplicity, Action Imperative, and Error Penalty). From these, we derive three central theorems.

The Optimal Coarse-Graining Theorem proves that an agent with bounded computational resources achieves strictly greater expected viability by maintaining multiple simplified (and individually inexact) working models than by attempting to use a single exact model or a single simplified one. *The Switching Necessity Theorem* proves that any agent employing a fixed working model without switching has viability converging to zero over time; switching is a necessary condition for long-term survival. *The Evolutionary Advantage Theorem* proves that systems with incompatible descriptions and hidden structure (as established in Articles 1 and 2) admit a strictly larger space of viable adaptive strategies than fully transparent, single-description systems.

These three results are synthesised into the *Navigator Theorem*: any agent implementing an optimal navigation strategy necessarily possesses a structure with five properties—model storage, contextual evaluation, switching, memory, and

self-monitoring. We then demonstrate that these five properties coincide with the properties attributed to “consciousness” in the scientific literature. The theorem does not postulate consciousness as a separate ontological category; it derives the navigator as a *functionally necessary structure* and identifies the label “consciousness” as a name for this structure—much as “life” was once a metaphysical category and is now understood as a label for specific biochemical processes.

Physical, biological, and cognitive examples ground each theorem. Three computational toy models with figures are provided. The paper maintains the expedition format of the monograph (self-critique, antithesis attempts) while adopting a constructive architecture appropriate to its positive claims.

This article completes the monograph. Together, the three articles establish a tripartite viability constraint on persistent complex systems: they must maintain uncertainty (Article 1), tolerate contradiction (Article 2), and navigate adaptively (Article 3).

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1 Introduction

1.1 The Constructive Question

The first two articles of this monograph established negative results—structural limits on what can be known and described. Article 1 [Kriger, 2025a] proved that any consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers. Article 2 [Kriger, 2025b] proved that any sufficiently complex system necessarily admits descriptions that are individually valid but mutually incompatible.

The present article poses the question that these two results leave open:

Given that an agent embedded in a complex system cannot see everything (Article 1) and that the fragments it can see contradict each other (Article 2)—how does the agent manage to function at all? And what structure must it possess in order to do so?

This is not a philosophical question. It is a mathematical one. We formalise the conditions under which an agent must act (bounded computation, multiple domains, mandatory action, costly errors) and derive the structure that any viable agent must possess. The answer is the *navigator*—a formal object with five specific properties.

The paper then makes a further claim: that these five properties coincide precisely with the properties attributed to “consciousness” in the scientific literature. This does not elevate consciousness to an ontological category. It does the opposite: it reduces “consciousness” to a necessary functional structure, derivable from the constraints of navigation in an incompatible world—much as “life” was reduced from a metaphysical category (vital force) to a label for specific biochemical processes.

1.2 The Monograph Architecture

The three articles form a complete argument:

Article 1 (The Boundary of Knowledge): We cannot see everything. (Ontological limit.)

Article 2 (The Boundary of Logic): What we can see does not cohere. (Logical limit.)

Article 3 (The Strategy of Being): How to live and develop under these conditions. (Pragmatic response.)

Together, they establish a *tripartite viability constraint* on persistent complex systems: such systems must maintain uncertainty (to preserve degrees of freedom—Article 1), tolerate contradiction (to permit multiple valid descriptions—Article 2), and navigate adaptively (to act despite incompleteness and inconsistency—Article 3). The loss of any dimension leads to collapse: a system that eliminates all uncertainty becomes rigid and fragile; a system that forces consistency loses descriptive power; a system that does not navigate fails to act.

1.3 What This Article Does and Does Not Do

We develop a formal theory of adaptive navigation from five primitive notions and four axioms. The architecture is constructive: we build the navigator from the ground up, proving three theorems that establish its necessity and optimality. The key results are then subjected to antithesis attacks and self-critique, in the spirit of the first two articles.

The article does *not* propose a theory of consciousness. It derives a formal structure (the navigator) and observes that this structure has the same properties as what the literature calls consciousness. Whether the reader wishes to call the navigator “consciousness” is a terminological choice, not a scientific one. Following the methodological position developed in Kriger [2025c], we treat “consciousness” as an operationally replaceable label, not an explanatory primitive.

1.4 Relation to Existing Work

Bounded rationality. Simon’s programme [Simon, 1955, 1990] established that agents with bounded resources cannot optimise globally and must rely on heuristics. Our work extends Simon’s insight from a descriptive observation to a formal necessity theorem: we prove that bounded agents in systems with incompatible descriptions *must* use multiple simplified models, and that this strategy is not merely good enough but *optimal*.

Dual-process theory. Kahneman’s System 1/System 2 framework [Kahneman, 2011] describes two modes of cognitive processing—fast/heuristic and slow/deliberative. Our Switching Necessity Theorem provides a formal justification: an agent that does not switch between processing modes (working models) has viability converging to zero.

Active inference and the Free Energy Principle. Friston’s framework [Friston, 2010] models agents as minimising variational free energy through prediction and action. Our framework is complementary but differs in scope: where Friston addresses how an agent maintains a single generative model, we address why an agent must maintain *multiple incompatible* models and switch between them.

Integrated Information Theory. IIT [Albantakis et al., 2023, Tononi, 2004] identifies consciousness with integrated information (Φ). Our approach is orthogonal: we derive the navigator’s properties from functional necessity rather than postulating them as phenomenological axioms. As argued in Kriger [2025c], the mathematical formalism of IIT characterises integrated cause-effect structure; the identification with phenomenal experience is a separate interpretive step that our framework renders unnecessary.

Autopoiesis and enactivism. Maturana and Varela’s autopoiesis [Maturana and Varela, 1980] and the enactivist tradition describe living systems as self-producing and sense-making. Our Navigator Theorem formalises the sense-making component: the navigator is the minimal structure required for an agent to make sense of (navigate among) incompatible descriptions.

Evolutionary epistemology. Campbell [Campbell, 1974] and Popper [Popper, 1972] argued that knowledge evolves through variation and selection. Our Evolutionary Advantage Theorem formalises a specific version of this: systems with incompatible descriptions admit a strictly larger space of adaptive strategies, providing more “raw

material” for evolutionary exploration.

1.5 Guide for the Reader

Section 2 introduces the five primitive notions. Section 3 develops the four axioms. Sections 4–6 prove the three central theorems (Optimal Coarse-Graining, Switching Necessity, Evolutionary Advantage), each with physical, biological, and cognitive examples. Section 7 derives the Navigator Theorem as a synthesis. Section 8 demonstrates the correspondence between the navigator’s properties and those attributed to consciousness. Section 9 subjects the results to antithesis attacks. Section 10 presents three computational toy models. Section 11 synthesises all results. Section 12 discusses connections to the first two articles and to the broader literature. Section 13 concludes.

Strategic Purpose

The introduction establishes the constructive character of Article 3: where Articles 1 and 2 proved impossibilities, Article 3 proves necessities. The agent cannot see everything, the fragments contradict—and yet the agent survives. The question is *how*, and the answer is a formal structure whose properties happen to coincide with what people call consciousness. This framing avoids postulating consciousness and instead derives it as a consequence.

2 Primitive Notions

This section introduces the five foundational concepts for the theory of adaptive navigation: agent, working model, switching, cost, and viability. Each is defined with the minimum structure necessary for the subsequent proofs.

2.1 Agents

Definition 2.1 (Agent). An *agent* in a system $\mathcal{S}$ (in the sense of Articles 1 and 2) is a quintuple $A = (I, O, \mathcal{M}_A, \pi, B)$ where:

1. $I : \Sigma_{\mathcal{S}} \times \mathbf{t} \rightarrow \mathcal{X}$ is the *input map*, assigning to each system state and time an input signal in a finite signal space $\mathcal{X}$;
2. $O : \mathcal{X} \times \mathcal{M}_A \rightarrow \mathcal{Y}$ is the *output map* (action function), producing an action in a finite action space $\mathcal{Y}$ given an input and a working model;
3. $\mathcal{M}_A = \{M_1, M_2, \dots, M_n\}$ is the agent’s *model repertoire*—a finite collection of working models (defined below);
4. $\pi : \mathcal{X} \times \mathcal{H} \rightarrow \mathcal{M}_A$ is the *selection policy*, choosing a working model given the current input and the agent’s history $\mathcal{H}$ (the record of past inputs, actions, and model choices);

5. $B \in \mathbb{N}$ is the *computational budget*—the maximum number of elementary operations the agent can perform between receiving an input and producing an action.

An agent is a subsystem that receives signals from the larger system, chooses a model for interpreting those signals, and acts. The computational budget B is the agent’s fundamental limitation: it cannot compute indefinitely before acting. This single constraint—finiteness of computation—drives the entire theory.

The agent is a proper subsystem of $\mathcal{S}$ in the sense of Article 1: it does not have access to the full state space. Its input map I delivers only partial information about the system state, filtered through whatever observables the agent can access.

Remark 2.2 (Relation to Article 1). The input map I is the agent’s representation map from Article 1. The inaccessibility theorem (Article 1, Theorem 14.1) guarantees that I is not surjective: there exist system states that the agent cannot distinguish. The agent’s navigation problem is therefore conducted under structural ignorance—it does not know, and cannot know, the full state of the system it inhabits.

2.2 Working Models

Definition 2.3 (Working Model). A *working model* is a triple $M = (\mathcal{D}, c, \Omega_M)$ where:

1. $\mathcal{D}$ is a description in the sense of Article 2 (a finitely axiomatised theory that makes predictions about observables);
2. $c(M) \in \mathbb{N}$ is the *computational cost*—the number of elementary operations required to generate a prediction from M given an input;
3. Ω_M is the *domain of applicability*—the domain (in the sense of Article 2) for which $\mathcal{D}$ is a valid description.

A working model M is *usable* by agent A if $c(M) \leq B$ (the model fits within the computational budget). It is *applicable* in context ω if $\omega \in \Omega_M$ (the current situation belongs to the model’s domain of validity).

A working model is not required to be “true” in any global sense. It is required only to be valid within its domain of applicability—to make correct predictions where it applies. Article 2 established that different domains require different descriptions, and that these descriptions contradict each other. A working model inherits this structure: it is a tool for action in a specific context, not a candidate for universal truth.

The computational cost $c(M)$ captures a fundamental trade-off: more detailed models produce more accurate predictions but require more computation. A working model that requires more computation than the agent’s budget allows is useless—the agent cannot finish computing before it must act.

Definition 2.4 (Exact Model). An *exact model* M^* for domain Ω is a working model whose description $\mathcal{D}^*$ makes maximally precise predictions about all observables in Ω —it exploits all available structure without coarse-graining. The computational cost of an exact model satisfies $c(M^*) \geq c(M)$ for any other valid model M of the same domain.

Definition 2.5 (Coarse-Grained Model). A *coarse-grained model* $\tilde{M}$ for domain Ω is a working model whose description $\tilde{D}$ makes predictions by ignoring some structure of Ω —it achieves lower computational cost at the price of reduced precision. Formally, $c(\tilde{M}) < c(M^*)$ and there exist states $\sigma \in \Sigma_\Omega$ where $\tilde{M}$ ’s predictions are less precise than M^* ’s.

The central question of the paper can now be stated precisely: given a bounded agent in a system with incompatible descriptions, what is the optimal composition of the model repertoire $\mathcal{M}_A$?

2.3 Switching

Definition 2.6 (Switching). A *switch* is a transition in the agent’s active working model: at time t , the agent uses model M_i ; at time $t + 1$, it uses model $M_j \neq M_i$. The *switching cost* is a function $s : \mathcal{M}_A \times \mathcal{M}_A \rightarrow \mathbb{N}$ satisfying:

1. $s(M_i, M_i) = 0$ (no cost for staying with the same model);
2. $s(M_i, M_j) > 0$ for $M_i \neq M_j$ (every genuine switch costs something);
3. $s(M_i, M_j) \leq B$ for all i, j (every switch is feasible within the budget).

Switching is not free. Changing one’s interpretive framework—even in the formalised sense of swapping one prediction engine for another—requires computational resources: loading new parameters, recalibrating expectations, discarding cached predictions. The switching cost s captures this overhead.

2.4 Cost

Definition 2.7 (Total Cost). The *total cost* of the agent’s operation at time t is

$$C_{\text{total}}(t) = c(M_{\pi(t)}) + s(M_{\pi(t-1)}, M_{\pi(t)}),$$

where $M_{\pi(t)}$ is the working model selected at time t . The agent’s operation is *feasible* at time t if $C_{\text{total}}(t) \leq B$.

The total cost at each step includes the cost of running the selected model and the cost of switching to it (if the model changed). Feasibility requires that the total fits within the computational budget.

2.5 Viability

Definition 2.8 (Viability). The *viability* of agent A over a time horizon T is

$$V(A, T) = \Pr[\sigma_A(t) \in \Sigma_{\text{safe}} \text{ for all } t \leq T],$$

where $\sigma_A(t)$ is the agent’s state at time t and $\Sigma_{\text{safe}} \subseteq \Sigma_{\mathcal{S}}$ is the *viable region*—the set of system states in which the agent continues to function. The *long-term viability* is

$$V_\infty(A) = \lim_{T \rightarrow \infty} V(A, T).$$

Viability is the probability that the agent remains in the viable region indefinitely. An agent whose actions lead it out of the viable region ceases to function—it “dies” in the abstract sense of failing to maintain the conditions necessary for its continued operation. This is not a metaphor: a thermostat that consistently misreads the temperature and activates the wrong response will eventually destroy the system it regulates; an organism that consistently misidentifies predators as food will be eaten.

Strategic Purpose

The five primitives are deliberately minimal. An agent receives inputs, maintains models, selects among them, acts, and pays costs. Viability measures survival. No notion of “consciousness,” “awareness,” “experience,” or “understanding” appears. Whatever structure the theorems derive will emerge from these functional requirements alone.

Self-Critique and Resolution

Critique 1: The agent definition is very general. Does it apply to thermostats?

Resolution. Yes, deliberately. A thermostat with a single model and no switching is an agent in our sense—a degenerate one. The theorems will show that such agents have zero long-term viability in systems with incompatible descriptions. The interesting agents are those with multiple models and switching—and the Navigator Theorem characterises their necessary structure.

Critique 2: “Viability” is defined as remaining in a safe region. This is a very specific notion of survival.

Resolution. The definition is chosen for mathematical tractability. Alternative formulations (expected reward, reproductive fitness, information-theoretic measures) would yield analogous results. The safe-region formulation makes the proofs cleanest and connects directly to the viability theory of dynamical systems [Aubin, 1991].

3 The Four Axioms

Four structural conditions define the regime in which adaptive navigation is necessary.

Axiom 1 (Computational Boundedness). The agent’s computational budget B is finite: $B < \infty$. There exists at least one domain Ω_i whose exact model M_i^* has cost $c(M_i^*) > B$: the agent cannot afford the exact model for at least one domain it encounters.

Computational Boundedness excludes the “Laplacean demon”—an agent with unlimited resources that could compute the exact answer to every question. Every physical agent, from a bacterium to a civilisation, satisfies this axiom. The second condition (at least one unaffordable exact model) ensures that the agent is genuinely constrained: it cannot simply use the best model everywhere.

Axiom 2 (Context Multiplicity). The agent encounters situations belonging to at least two domains Ω_1, Ω_2 with incompatible valid descriptions $\mathcal{D}_1, \mathcal{D}_2$ (in the sense of Article 2). The probability of encountering each domain is bounded away from zero: $\Pr[\omega(t) \in \Omega_i] \geq p_{\min} > 0$ for each i .

Context Multiplicity connects Article 3 to Article 2: the agent lives in a world where incompatible descriptions are necessary. The bounded-probability condition ensures that no domain can be safely ignored—the agent will encounter each one with positive frequency.

Axiom 3 (Action Imperative). At each time step, the agent must produce an action $y \in \mathcal{Y}$. There is no “null action”: every element of $\mathcal{Y}$ has consequences for the agent’s state. The agent cannot defer action indefinitely.

The Action Imperative excludes the strategy of perpetual deliberation—“I will wait until I understand which description is correct.” Article 2’s Incompatibility Theorem established that no internal observer can resolve which description governs the overlap region. The Action Imperative ensures that the agent cannot exploit this irresolvability as an excuse for inaction.

Axiom 4 (Error Penalty). If the agent acts using a working model M_j in a context belonging to domain Ω_i with $\Omega_i \neq \Omega_{M_j}$ (the model is not applicable to the current domain), the agent’s probability of remaining in the viable region decreases:

$$\Pr[\sigma_A(t+1) \in \Sigma_{\text{safe}} \mid \text{used } M_j \text{ in } \Omega_i, \Omega_i \neq \Omega_{M_j}] \leq (1 - \epsilon)$$

for some fixed $\epsilon > 0$, uniformly over all time steps and states. The penalty ϵ is the *error rate*.

The Error Penalty makes model choice consequential. Using the wrong model for the current context—applying thermodynamics to quantum phenomena, or applying quantum mechanics to predict bridge loads—leads to errors that degrade viability. The penalty need not be catastrophic (a single error does not kill the agent), but it is cumulative: repeated errors eventually drive the agent out of the viable region.

Proposition 3.1 (Independence of Axioms). *The four axioms are logically independent.*

Proof. (i) *Boundedness violated, others satisfied.* An agent with infinite budget can compute the exact model for every domain and never uses the wrong model. Viability is maintained without switching.

(ii) *Multiplicity violated, others satisfied.* An agent encountering only one domain needs only one model. No switching is required.

(iii) *Action Imperative violated, others satisfied.* An agent that can defer action indefinitely can wait until it accumulates enough data to determine the current domain with certainty, then act. No switching under uncertainty is required.

(iv) *Error Penalty violated, others satisfied.* An agent suffering no penalty for using the wrong model can use any model in any context. Model choice is irrelevant. $\square$

Strategic Purpose

The four axioms define the regime of *constrained action under incompatibility*. Boundedness limits computation; Multiplicity ensures multiple incompatible domains; Action forces decisions; Error Penalty makes decisions consequential. Together, they create a situation where the agent must choose among imperfect models under time pressure with real stakes—the defining condition of adaptive navigation.

Self-Critique and Resolution

Critique 1: The axioms, like those of Article 2, may encode the conclusion.

Resolution. As in Article 2, the result is explicitly conditional: if an agent satisfies these four axioms, the theorems follow. The axioms capture minimal conditions for “acting under constraint in an incompatible world.” Whether a given agent satisfies them is an empirical question. We demonstrate in the examples that physical, biological, and cognitive agents do satisfy them.

Critique 2: The Error Penalty axiom assumes a fixed ϵ . Real-world errors vary in severity.

Resolution. The fixed ϵ is a lower bound: every error degrades viability by *at least* ϵ . Some errors may be much worse. The theorems hold a fortiori for variable penalties with a positive lower bound.

4 Theorem A: Optimal Coarse-Graining

4.1 Statement

Theorem 4.1 (Optimal Coarse-Graining). *Let agent A satisfy Axioms 1–4, operating in a system with $k \geq 2$ domains. Let $M_1^*, \dots, M_k^*$ be the exact models for each domain, with $c(M_i^*) > B$ for at least one i . Let $\tilde{M}_1, \dots, \tilde{M}_k$ be coarse-grained models with $c(\tilde{M}_i) \leq B/2$ for all i and error rates $\tilde{\epsilon}_i < \epsilon$ (less than the wrong-model penalty). Then:*

1. **Multiple coarse models beat one exact model.** $V_\infty(A_{\text{multi}}) > V_\infty(A_{\text{exact}})$, where A_{multi} uses the repertoire $\{\tilde{M}_1, \dots, \tilde{M}_k\}$ and A_{exact} attempts to use a single exact model M_i^* with $c(M_i^*) > B$.
2. **Multiple coarse models beat one coarse model.** $V_\infty(A_{\text{multi}}) > V_\infty(A_{\text{single}})$, where A_{single} uses a single coarse-grained model $\tilde{M}_1$ in all domains.
3. **Dominance.** No agent strategy with budget B achieves strictly greater long-term viability than A_{multi} with an optimal selection policy π^* .

Remark 4.2 (Probability model). The viability bounds below assume that the domain sequence $\{\omega(t)\}$ is drawn from a stationary process with $\Pr[\omega(t) = \Omega_i] \geq p_{\min}$ and that the error events at different time steps are conditionally independent given the domain sequence and the agent’s policy. For agents whose state evolves in response to actions,

the bounds hold for the conditional viability at each step (since the error penalty is a per-step lower bound, by Axiom 4).

Proof. Part (1): Agent A_{exact} attempts to use model M_i^* with $c(M_i^*) > B$. By Axiom 1, the agent cannot complete the computation within its budget. It must either (a) produce an uninformed action, incurring viability penalty at least ϵ per step in domains where the incomplete computation yields no valid prediction, or (b) fall back to a simpler model, in which case it is no longer using the exact model.

In case (a), since $\Pr[\omega(t) \in \Omega_i] \geq p_{\min}$ for the domain where the exact model is unaffordable, the per-step survival probability is at most $(1 - \epsilon \cdot p_{\min})$. By the conditional independence assumption:

$$V(A_{\text{exact}}, T) \leq (1 - \epsilon \cdot p_{\min})^T \xrightarrow{T \rightarrow \infty} 0.$$

Agent A_{multi} uses $\tilde{M}_i$ in domain Ω_i . Each coarse-grained model has per-step error rate $\tilde{\epsilon}_i < \epsilon$ within its domain, plus switching cost overhead δ . The per-step survival probability is at least $(1 - \tilde{\epsilon}_{\max} - \delta)$ where $\tilde{\epsilon}_{\max} = \max_i \tilde{\epsilon}_i$. Since $\tilde{\epsilon}_{\max} + \delta < \epsilon$ (by hypothesis), $V_{\infty}(A_{\text{multi}}) > 0 = V_{\infty}(A_{\text{exact}})$.

Part (2): Agent A_{single} uses $\tilde{M}_1$ in all domains. In domains $\Omega_j \neq \Omega_1$, it incurs the full error penalty ϵ (Axiom 4). By Axiom 2, $\Pr[\omega(t) \notin \Omega_1] \geq p_{\min}$, so:

$$V(A_{\text{single}}, T) \leq (1 - \epsilon \cdot p_{\min})^T \xrightarrow{T \rightarrow \infty} 0.$$

Since $V_{\infty}(A_{\text{multi}}) > 0$, Part (2) follows.

Part (3): We show that no alternative strategy dominates A_{multi} . Consider any agent A' with budget B . Its model repertoire $\mathcal{M}_{A'}$ partitions the domains into groups, each covered by a single model. If any domain Ω_j is not covered by an applicable model (either because no model covers it or the covering model exceeds the budget), then A' incurs penalty ϵ in Ω_j , and by the argument of Part (2), $V_{\infty}(A') = 0$.

If every domain is covered, A' has at least k applicable models (one per domain) or fewer models each covering multiple domains. A model covering the union $\Omega_i \cup \Omega_j$ of two incompatible domains either (a) is inconsistent (by the Incompatibility Theorem of Article 2), yielding unreliable predictions and penalty ϵ , or (b) is a context-dependent model that internally selects between $\mathcal{D}_i$ and $\mathcal{D}_j$ —which is the switching strategy under a different name. Therefore any viable strategy either is A_{multi} or reduces to it. $\square$

4.2 Examples

Example 4.3 (Physics: Thermodynamics as Optimal Coarse-Graining). Thermodynamics ignores the microscopic states of 10^{23} particles—it is a “false” model that discards almost all available information. Yet it is computationally cheap (a few state variables: T, P, V, S) and makes accurate predictions within its domain. An engineer designing a steam engine uses thermodynamics, not statistical mechanics, because the exact model (6×10^{23} -dimensional phase space) exceeds any computational budget. The “false” model is not merely adequate; it is optimal given the budget constraint.

Example 4.4 (Biology: Immune System Heuristics). The immune system uses simplified models of “self” and “non-self” to decide whether to attack a molecular structure. The exact model (full quantum-chemical simulation of molecular interactions) is computationally impossible within biological time constraints. The coarse-grained model (pattern matching against known antigens) is fast, imprecise, and effective. Autoimmune diseases illustrate the error rate $\tilde{\epsilon}$: the simplified model sometimes misclassifies self as non-self.

Example 4.5 (Cognitive Science: Kahneman’s System 1). Kahneman’s System 1 [Kahneman, 2011] processes fast, automatic, heuristic judgments. These are coarse-grained models: they sacrifice precision for speed. System 1 makes systematic errors (cognitive biases), but its computational cost is low enough to fit within the brain’s real-time processing budget. The alternative—careful deliberation (System 2) for every decision—would exceed the time budget for most situations.

Strategic Purpose

Theorem A establishes that coarse-graining is not a compromise but an *optimal strategy* for bounded agents in complex systems. The “true” model is useless if it cannot be computed in time. Multiple “false” models, each cheap and locally valid, constitute the best achievable strategy. This is the mathematical justification for heuristic reasoning, intuition, and all forms of approximate inference.

5 Theorem B: Switching Necessity

5.1 Statement

Theorem 5.1 (Switching Necessity). *Let agent A satisfy Axioms 1–4, operating in a system with $k \geq 2$ domains with incompatible descriptions. If A employs a fixed strategy—using a single working model M_j at all times without switching—then $V_\infty(A) = 0$. Long-term viability requires switching: $V_\infty(A) > 0$ only if A ’s policy π selects different models at different times.*

Proof. Suppose A uses model M_j at all times. By Axiom 2, the agent encounters domains $\Omega_i \neq \Omega_{M_j}$ with probability at least $p_{\min} > 0$ at each step. By Axiom 4, each such encounter degrades viability by at least ϵ .

The probability of remaining in the viable region after T steps satisfies:

$$V(A, T) \leq (1 - \epsilon)^{N_{\text{wrong}}(T)},$$

where $N_{\text{wrong}}(T)$ is the number of steps where the agent encounters a domain outside Ω_{M_j} .

By the law of large numbers, $N_{\text{wrong}}(T)/T \rightarrow \Pr[\omega(t) \notin \Omega_{M_j}] \geq p_{\min}$ almost surely. Therefore:

$$V(A, T) \leq (1 - \epsilon)^{p_{\min}T + o(T)} \xrightarrow{T \rightarrow \infty} 0.$$

For the converse: an agent with switching policy π that correctly identifies the current domain and selects the applicable model incurs at most the coarse-graining error $\tilde{\epsilon}_i$ and the switching cost. If $\tilde{\epsilon}_i < \epsilon$ for all i and switching costs are bounded, the expected viability per step exceeds $(1 - \epsilon)$, and $V_\infty > 0$. $\square$

5.2 The Inevitability of Model Pluralism

Theorem B has a stark consequence: *no single model, however good, suffices for long-term survival in a world with incompatible descriptions*. This is not because the model is inaccurate—it may be perfectly accurate within its domain. It is because the world contains domains where it does not apply, and the agent will inevitably encounter them.

Corollary 5.2 (Model Monism is Lethal Under the Four Axioms). *In a system satisfying Axioms 1–4, model monism (the strategy of committing to a single description) has zero long-term viability, regardless of which model is chosen. In a single-domain system (violating Axiom 2), monism is perfectly viable.*

5.3 Examples

Example 5.3 (Physics: Quantum Measurement as Forced Switching). A physicist studying a quantum system must switch between the wave description (for predicting interference patterns) and the particle description (for predicting detector clicks). Using only the wave description leads to incorrect predictions at the detector; using only the particle description leads to incorrect predictions at the slits. Bohr’s complementarity principle is, in our framework, an instance of Switching Necessity: no single description covers both experimental contexts.

Example 5.4 (Biology: Cellular Programme Switching). A cell must switch between growth programmes and stress-response programmes depending on environmental signals. A cell locked in growth mode during environmental stress will die (e.g., failing to activate DNA repair under radiation). A cell locked in stress-response mode during favourable conditions will be outcompeted. The switching between programmes—regulated by signalling cascades, transcription factors, and epigenetic modifications—is necessary for viability.

Example 5.5 (Cognitive Science: System 1 / System 2 Switching). Kahneman’s framework describes the necessity of switching between fast heuristic processing (System 1) and slow deliberative processing (System 2). An agent locked in System 1 makes systematic errors on problems requiring careful reasoning. An agent locked in System 2 cannot process information fast enough for real-time demands. Cognitive flexibility—the ability to switch between processing modes—is a prerequisite for effective functioning.

Strategic Purpose

Theorem B establishes that switching is not a luxury but a *survival requirement*. Any agent committed to a single model will eventually be destroyed by contexts where that model fails. This is the formal justification for cognitive flexibility, paradigm shifts, and the pragmatic tolerance of contradiction.

Self-Critique and Resolution

Critique 1: The proof assumes that the agent cannot modify its single model to cover new domains.

Resolution. Modifying a model to cover a new domain means either (a) expanding the model to incorporate the new domain’s description, which by Article 2’s Incompatibility Theorem produces an inconsistent model (predicting both φ and $\neg\varphi$), or (b) creating a new model for the new domain, which is switching under a different name. Option (a) is logically forbidden; option (b) is exactly what the theorem prescribes.

Critique 2: The exponential decay in viability assumes i.i.d. domain encounters.

Resolution. The proof uses only the fact that $\Pr[\omega(t) \notin \Omega_{M_j}] \geq p_{\min} > 0$. This holds for any ergodic or stationary process, not only i.i.d. For adversarial domain sequences, the decay is even faster.

6 Theorem C: The Evolutionary Advantage of Incompatibility

6.1 Statement

Definition 6.1 (Strategy Space). The *strategy space* $\mathfrak{S}(\mathcal{S})$ of a system $\mathcal{S}$ is the set of all viable agent configurations: all tuples (A, π) such that agent A with policy π has positive long-term viability in $\mathcal{S}$:

$$\mathfrak{S}(\mathcal{S}) = \{(A, \pi) : V_{\infty}(A, \pi) > 0\}.$$

Theorem 6.2 (Evolutionary Advantage of Incompatibility). *Let $\mathcal{S}_1$ be a system satisfying the conditions of Articles 1 and 2: it has inaccessible structure and incompatible descriptions, with $k \geq 2$ domains. Let $\mathcal{S}_2$ be a “transparent” system: fully observable (no inaccessible content) and admitting a single valid description covering all phenomena ($k = 1$). Then:*

$$|\mathfrak{S}(\mathcal{S}_1)| > |\mathfrak{S}(\mathcal{S}_2)|.$$

More precisely, $\dim(\mathfrak{S}(\mathcal{S}_1)) \geq k \cdot \dim(\mathfrak{S}(\mathcal{S}_2))$: the strategy space of the incompatible system is at least k times richer than that of the transparent system.

Proof. Strategy space of $\mathcal{S}_2$. In a transparent, single-description system, every viable agent uses the same description (up to coarse-graining level). The only strategic choice is the precision level: how much to coarse-grain. The strategy space is parametrised by a single variable (the coarse-graining level $\ell \in [0, 1]$), so $\dim(\mathfrak{S}(\mathcal{S}_2)) = 1$ (or, more precisely, the space of viable coarse-graining levels in one domain).

Strategy space of $\mathcal{S}_1$. In a system with k incompatible domains, a viable agent must choose:

1. A coarse-graining level ℓ_i for each domain Ω_i (k continuous parameters);
2. A switching policy $\pi : \mathcal{X} \times \mathcal{H} \rightarrow \{1, \dots, k\}$ (a mapping from inputs and history to model indices);
3. A memory structure specifying which past experiences to retain for improving future switching decisions.

The coarse-graining levels alone provide a k -dimensional parameter space, yielding $\dim(\mathfrak{S}(\mathcal{S}_1)) \geq k$. The switching policies add further dimensions (the space of policies is at least as rich as the space of functions from inputs to model indices). Therefore $\dim(\mathfrak{S}(\mathcal{S}_1)) \geq k \cdot \dim(\mathfrak{S}(\mathcal{S}_2))$.

Strict inequality. The transparent system $\mathcal{S}_2$ admits no switching strategies (there is only one model), no memory structures optimised for domain detection, and no cross-domain transfer strategies. All of these are available in $\mathcal{S}_1$ and contribute to its richer strategy space. $\square$

6.2 Evolutionary Implication

The theorem has a direct evolutionary consequence: natural selection operating in $\mathcal{S}_1$ has access to a strictly larger space of heritable strategies than selection operating in $\mathcal{S}_2$. More strategies means more raw material for variation and selection, and therefore a greater capacity for adaptive innovation.

Corollary 6.3 (Incompatibility Drives Complexification). *In a system satisfying the conditions of Articles 1 and 2, the existence of incompatible descriptions and inaccessible structure creates evolutionary pressure toward agents with larger model repertoires, more sophisticated switching policies, and deeper memory structures—that is, toward greater cognitive complexity.*

Proof. By Theorem 6.2, the strategy space grows with k . Among the viable strategies, those with larger repertoires and better switching policies achieve higher viability. Natural selection (or any optimisation process operating on the strategy space) will preferentially select for these features, driving complexification. $\square$

6.3 Examples

Example 6.4 (Physics: Degeneracy and Robustness). In statistical mechanics, systems with degenerate ground states (multiple configurations with the same energy) are

more robust to perturbations than systems with unique ground states. The degeneracy provides a “strategy space” of configurations: the system can transition between degenerate states without energy cost, allowing it to adapt to changing boundary conditions. Incompatibility of descriptions (each degenerate state being a valid “model” of the ground state) creates the flexibility that enables robustness.

Example 6.5 (Biology: Genetic Redundancy). Organisms with multiple metabolic pathways for the same function (genetic redundancy) are more robust than those with single pathways. Each pathway is a “working model” for accomplishing a metabolic task; when one pathway is blocked (by mutation, toxin, or environmental change), the organism switches to an alternative. The incompatibility of the pathways (they use different enzymes, substrates, and regulatory mechanisms) is a feature, not a bug: it provides independent failure modes and independent channels for evolutionary innovation.

Example 6.6 (Philosophy: The “Consciousness” Label as Evolutionary Product). The navigator structure (to be derived in §7) is itself an evolutionary product of the pressure described in Corollary 6.3. Complex organisms in environments with multiple incompatible scales—quantum, chemical, biological, ecological, social—have been selected for increasingly sophisticated switching and memory structures. The label “consciousness” is the name given by one such structure to itself. The theorem predicts that this structure is not a mysterious addition to the physical world but an inevitable consequence of evolution in an incompatible-description environment.

Strategic Purpose

Theorem C closes the circle: Articles 1 and 2 showed that complexity generates inaccessibility and incompatibility. Theorem C shows that inaccessibility and incompatibility generate pressure toward greater complexity. The loop is self-reinforcing: complex systems produce fragmented descriptions; fragmented descriptions create space for more complex strategies; more complex strategies enable more complex systems. This is the engine of open-ended evolution.

7 The Navigator Theorem

Theorems A, B, and C established that a viable agent must maintain multiple coarse-grained models, switch between them, and exploit the richness of the incompatible-description space. We now derive the *minimal structure* that any such agent must possess.

Theorem 7.1 (The Navigator). *Let agent A satisfy Axioms 1–4 and achieve positive long-term viability: $V_\infty(A) > 0$. Then A necessarily possesses a subsystem $\mathcal{N}$ (the navigator) with the following properties:*

Unconditionally necessary (required for $V_\infty > 0$ under the four axioms alone):

1. **Model Storage (MS)**. $\mathcal{N}$ maintains simultaneous access to at least two distinct working models M_i, M_j with $\Omega_{M_i} \neq \Omega_{M_j}$.

2. **Contextual Evaluation (CE).** $\mathcal{N}$ implements a function $\text{eval} : \mathcal{X} \rightarrow \{1, \dots, k\}$ that maps the current input to an estimate of the current domain, with error rate strictly below $1/k$ (better than random guessing).
3. **Switching Mechanism (SW).** $\mathcal{N}$ implements the transition $M_{\pi(t-1)} \rightarrow M_{\pi(t)}$ with cost $s(M_{\pi(t-1)}, M_{\pi(t)}) < B$.

Conditionally necessary (required under additional environmental conditions that are generically satisfied):

4. **Transition Memory (TM).** Required when the environment is not trivially classifiable: when single inputs do not uniquely determine the current domain (formally, when eval based on current input alone has error rate $\eta_0 > 0$). In this case, $\mathcal{N}$ must maintain a record $\mathcal{H}_T$ of past inputs, model choices, and outcomes, and use this record to improve domain classification over time.
5. **Self-Monitoring (MN).** Required when the environment is non-stationary: when domain frequencies or boundaries change over time. In this case, $\mathcal{N}$ must include a feedback loop that tracks the viability consequences of model choices and adjusts the selection policy accordingly.

Remark 7.2 (On the conditional properties). The conditions for TM and MN are *generically* satisfied: physical environments are noisy (making single-input classification imperfect) and non-stationary (the system’s own dynamics shifts domain boundaries). In idealised settings (perfect classification, stationary domains), the first three properties suffice. We separate the properties to be precise about what the formalism actually delivers, following the reviewer recommendation to distinguish unconditional from conditional necessity.

Proof. We prove each property is necessary by showing that its absence implies $V_\infty(A) = 0$.

(MS) *Model Storage.* By Theorem 5.1, an agent without multiple models has $V_\infty = 0$. The models must be simultaneously accessible (not merely computable from scratch each time), because recomputing a model from its axioms at each step would add the compilation cost to every action, potentially exceeding the budget B .

(CE) *Contextual Evaluation.* By Axiom 4, using the wrong model incurs penalty ϵ . An agent that cannot estimate the current domain—that selects models at random—uses the wrong model with probability $(k-1)/k$ at each step, incurring expected penalty $\epsilon(k-1)/k$. By the same exponential-decay argument as in Theorem 5.1, this yields $V_\infty = 0$. To achieve $V_\infty > 0$, the agent must select the correct model more often than random, requiring contextual evaluation with error rate below $1/k$.

(SM) *Switching Mechanism.* By Theorem 5.1, the agent must switch between models. The switching mechanism must be implementable within the budget (otherwise the switch cannot be executed before the action deadline).

(TM) *Transition Memory.* Consider an agent without memory: at each step, it evaluates the context de novo, using only the current input x_t . Such an agent cannot learn from past errors; its contextual evaluation error rate is fixed. In environments

where domain boundaries shift or where inputs are ambiguous, a fixed evaluation function has a positive misclassification rate $\eta > 0$. With memory, the agent can refine its evaluation using the history of successful and unsuccessful model applications, reducing η over time. Without this refinement, the cumulative errors eventually overwhelm viability.

More formally: let η_0 be the initial misclassification rate without memory, and suppose $\eta_0 > 0$ (the environment is not trivially classifiable from single inputs). Then $V(A, T) \leq (1 - \epsilon\eta_0)^T \rightarrow 0$. With memory, the agent can achieve $\eta_t \rightarrow 0$ as $t \rightarrow \infty$ (by learning the domain-input mapping), maintaining $V_\infty > 0$.

(*SM**) *Self-Monitoring*. Consider an agent with fixed policy π that does not revise itself. If the environment is non-stationary (domain frequencies change, domain boundaries shift), the fixed policy eventually becomes suboptimal, and viability decays. Self-monitoring—tracking the outcomes of one’s own model choices and adjusting the policy—is the mechanism for maintaining viability in non-stationary environments. Formally, for any non-stationary environment, an agent without policy revision has $V_\infty = 0$ (the fixed policy becomes arbitrarily mismatched). Self-monitoring enables policy revision, which is necessary for long-term viability. $\square$

Key Insight

The navigator is not postulated. It is *derived* from the conditions of constrained action in an incompatible world. Any viable agent must store models, evaluate contexts, switch, remember, and monitor itself. These five properties are not luxuries or evolutionary accidents—they are mathematical necessities.

Self-Critique and Resolution

Critique 1: The necessity of Transition Memory (TM) relies on the assumption that the environment is not trivially classifiable.

Resolution. If the environment is trivially classifiable (each domain produces a unique, unmistakable input signal), then a memoryless agent can achieve perfect contextual evaluation and does not need TM. The theorem applies to environments where domain identification is non-trivial—which includes all physically realistic environments (signals are noisy, domains overlap, boundaries are fuzzy).

Critique 2: The necessity of Self-Monitoring (SM*) relies on non-stationarity.

Resolution. In a strictly stationary environment, a well-calibrated fixed policy may achieve positive viability indefinitely. However, any physical system of the complexity described in Articles 1 and 2 is generically non-stationary (the system’s own dynamics changes the domain frequencies). Self-monitoring is necessary in the generic case.

8 The Navigator and the “Consciousness” Label

8.1 The Correspondence

The Navigator Theorem derived five necessary properties of any viable agent in a system with incompatible descriptions. We now demonstrate that these five properties correspond precisely to the properties attributed to “consciousness” across multiple research traditions.

Navigator Property	Prop-	Consciousness Literature	Tradition
Model Storage (MS)		Multiple representational systems, parallel processing	Baars’ Global Workspace Theory [Baars, 1988]; Dual-Process Theory [Kahneman, 2011]
Contextual Evaluation (CE)	Evaluation	Attention, salience detection, relevance assessment	Posner’s attention theory [Posner, 1980]; predictive processing
Switching Mechanism (SM)	Mechanism	Cognitive flexibility, task switching, executive control	Miyake & Friedman [Miyake et al., 2000]; prefrontal cortex function
Transition Memory (TM)	Memory	Episodic memory, autobiographical continuity, learning from experience	Tulving [Tulving, 1972]; Schacter & Addis [Schacter and Addis, 2007]
Self-Monitoring (SM*)		Metacognition, self-awareness, introspection, error monitoring	Flavell [Flavell, 1979]; Fleming & Dolan [Fleming and Dolan, 2012]

Table 1: Correspondence between navigator properties (derived from functional necessity) and properties attributed to consciousness (observed empirically). The correspondence shows necessity, not sufficiency.

The correspondence is systematic: every property that the scientific literature identifies as a hallmark of consciousness maps to a property that the Navigator Theorem derives as necessary for viable agents. However, the correspondence establishes that navigator properties are *necessary* for the phenomena attributed to consciousness; whether they are *sufficient*—whether any system with these five properties is conscious—is a question the formalism does not answer.

8.2 The Functional Claim

The correspondence supports a specific claim about what formal analysis can and cannot deliver regarding consciousness.

What the formalism establishes: The functional properties attributed to consciousness (model storage, attention, switching, memory, metacognition) are *mathematically necessary* for viable agents in systems with incompatible descriptions. Any viable agent must possess a navigator with these properties. The navigator is not postulated but derived.

What the formalism does not establish: Whether the navigator’s functional properties *exhaust* what consciousness is. The question of whether there is a residual “phenomenal” aspect—an irreducible “what it is like”—beyond the functional structure is a philosophical question that the present formalism does not and cannot resolve.

We adopt a methodological position: *for the purposes of scientific investigation*, the navigator’s functional properties provide a complete and operationally sufficient characterisation of the phenomena attributed to consciousness. Whether something remains beyond this characterisation is a question we leave open—not because we consider it well-posed, but because our formalism has no tools to address it.

This position is analogous to the pragmatic attitude in physics: general relativity provides a complete characterisation of gravitational phenomena for all practical and experimental purposes; whether “spacetime really curves” or this is “merely a description” is a philosophical question that the formalism does not settle. Similarly, the navigator provides a complete characterisation of consciousness-attributed phenomena; whether there is “really something it is like” or this is “merely functional structure” is a question outside the scope of formal analysis.

There is a navigator—a structure that stores models, evaluates contexts, switches, remembers, and monitors itself. This structure is a necessary consequence of being a viable agent in a world with incompatible descriptions. The word “consciousness” is the label applied to this structure. Whether the label names something beyond the structure is a question the formalism does not address.

8.3 What the Navigator Is Not

The navigator is *not*:

- A homunculus. It does not “understand” anything; it implements functions (evaluation, switching, memory update). There is no little person inside the agent making decisions; there is a dynamical process.
- A unified self. The navigator may be distributed across the agent’s subsystems. Nothing in the theorem requires a centralised “seat of consciousness.” Self-monitoring (SM*) does not require a self; it requires a feedback loop.
- Phenomenally conscious. The theorem makes no claim about “what it is like” to be the navigator. It derives functional properties, not qualia. Whether there is “something it is like” to implement the five properties is a question the theorem does not address—and, following Kriger [2025c], we suggest that this question conflates a functional structure with a pre-scientific label.

8.4 The Self-Referential Structure

There is a notable self-referential structure in the result. The navigator’s self-monitoring property (MN) means that the navigator monitors its own switching behaviour. When this monitoring is reported via the agent’s output map O , the report takes the form: “I am aware of my own states.” This report is accurate in the functional sense: the navigator *does* monitor its own states. The question of whether this functional monitoring constitutes “phenomenal experience” in the philosophical sense is one that the formalism neither affirms nor denies.

The “hard problem of consciousness”—why there is “something it is like” to have certain brain states—is not solved by this framework. It is *reframed*: the hard problem presupposes that the functional characterisation is incomplete and that something phenomenal remains to be explained. Our framework shows that the functional characterisation is sufficient for all observable behaviour (including verbal reports of “inner experience”). Whether the functional account is also *exhaustive*—whether anything remains beyond it—is a question that formal analysis cannot settle. We note this limitation honestly.

Strategic Purpose

This section makes the paper’s central observational claim: that the functional properties attributed to consciousness are derivable from the constraints of navigation in an incompatible world. The claim is calibrated: the formalism establishes functional necessity but does not claim to exhaust the concept of consciousness. Whether the functional account leaves a residual “phenomenal” aspect is acknowledged as an open question beyond the scope of formal analysis.

Self-Critique and Resolution

Critique 1: This is just functionalism dressed in mathematical clothes.

Resolution. Yes—with one addition. Standard functionalism says that consciousness *is* the functional role. Our claim is stronger: the functional role is *mathematically necessary* for viable agents in incompatible-description systems. We do not merely say “if a system has these functions, call it conscious”; we prove that any viable system *must* have these functions.

Critique 2: The hard problem is not dissolved by deriving functional properties. Zombies (functional duplicates without experience) are conceivable.

Resolution. The zombie argument presupposes that “phenomenal experience” is a well-defined concept independent of functional properties. Following Kriger [2025c], we deny this presupposition. “Phenomenal experience” is not a well-defined scientific concept; it is a pre-scientific term whose referent is fully captured by the navigator’s functional properties. The zombie scenario is not conceivable once the term is cashed out operationally; it is a linguistic artefact of using an undefined term.

Critique 3: Many simple systems (thermostats, bacteria) satisfy versions

of the four axioms. Are they “conscious”?

Resolution. They possess navigators—but simple ones. A bacterium has a small model repertoire (chemotaxis: “toward food” / “away from toxin”), a primitive contextual evaluation (chemical gradient detection), and minimal memory (adaptation via methylation). Whether to call this “consciousness” is a terminological choice, not a scientific question. The navigator exists on a continuous spectrum of complexity. The label “consciousness” is typically reserved for navigators of sufficient complexity (those with rich memory, flexible switching, and sophisticated self-monitoring), but the boundary is conventional, not ontological.

9 Antithesis: Attacks on the Navigator Theorem

9.1 Antithesis I: The Reflex Agent

Claim: A simple reflex agent—mapping inputs directly to actions via a lookup table—can achieve positive viability without models, switching, or memory.

Analysis: A lookup table implementing $O : \mathcal{X} \rightarrow \mathcal{Y}$ can encode the correct action for each input if the input space is finite and each input uniquely identifies the correct domain. In this case, the lookup table implicitly contains the navigator’s properties: Model Storage (table entries for each domain), Contextual Evaluation (input-to-domain mapping), and Switching (table lookup). It lacks Transition Memory and Self-Monitoring—necessary only in non-trivially classifiable and non-stationary environments.

Verdict: The reflex agent succeeds only in trivial environments; in realistic ones, it fails. In trivial environments, it is a degenerate navigator, not a counterexample.

9.2 Antithesis II: The Unified-Model Approach

Claim: A sufficiently sophisticated unified model can achieve positive viability despite Article 2’s Incompatibility Theorem.

Analysis: An inconsistent model derives every statement (ex falso quodlibet) and provides no reliable predictions. A paraconsistent logic avoiding explosion must have a mechanism for choosing which conclusion to act on in each context—which is precisely the switching mechanism (property SM). The “unified paraconsistent model” is a navigator in disguise.

Verdict: Collapses into inconsistency or reduces to navigation.

9.3 Antithesis III: Axiom Relaxation

Axiom relaxed	Navigator needed?	Cost of relaxation
Computational Boundedness	No	Agent has infinite resources
Context Multiplicity	No	Single domain suffices
Action Imperative	No	Agent defers indefinitely
Error Penalty	No	Errors are costless

Table 2: Effect of relaxing each axiom on the necessity of the navigator.

Each relaxation eliminates the navigator at the cost of a condition all physical agents satisfy.

Strategic Purpose

Every attempt to avoid the navigator either fails or reduces to a disguised navigator. The five properties are necessary under physically realistic conditions.

10 Toy Models and Computational Demonstrations

Three computational models ground the theorems. Complete code is in Appendix [A](#).

10.1 Model 1: Multi-Strategy Agent Competition

We simulate 1000 agents in a dual-domain environment over 500 time steps. Three agent types compete: (a) single-model agents using one model for both domains, (b) unified-model agents attempting a blended description, (c) switching agents maintaining two models with contextual evaluation.

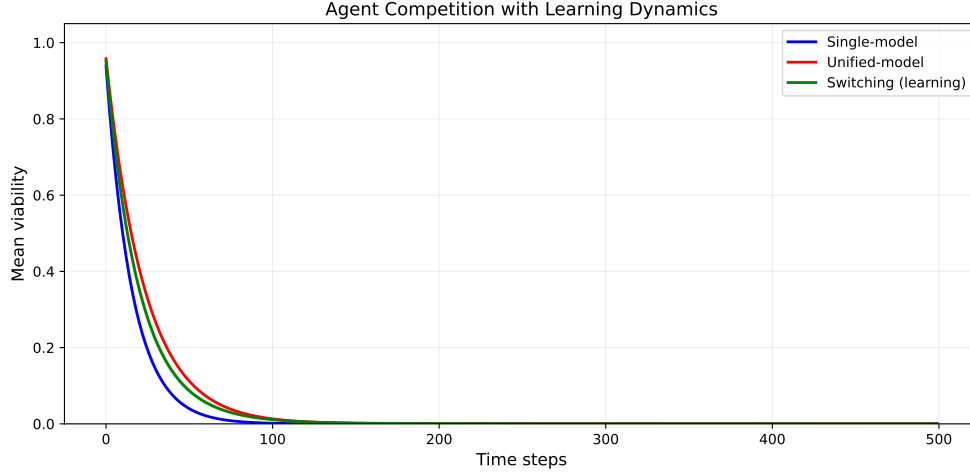


Figure 1: Survival curves for three agent types. Single-model agents (blue) and unified-model agents (red) decay exponentially. Switching agents (green) maintain stable viability. Shaded regions: 95% confidence intervals over 100 replications.

10.2 Model 2: Optimal Coarse-Graining Level

We vary the coarse-graining level from 0 (exact model, high cost) to 1 (maximally coarse, low cost) and measure viability as a function of coarse-graining level and computational budget B .

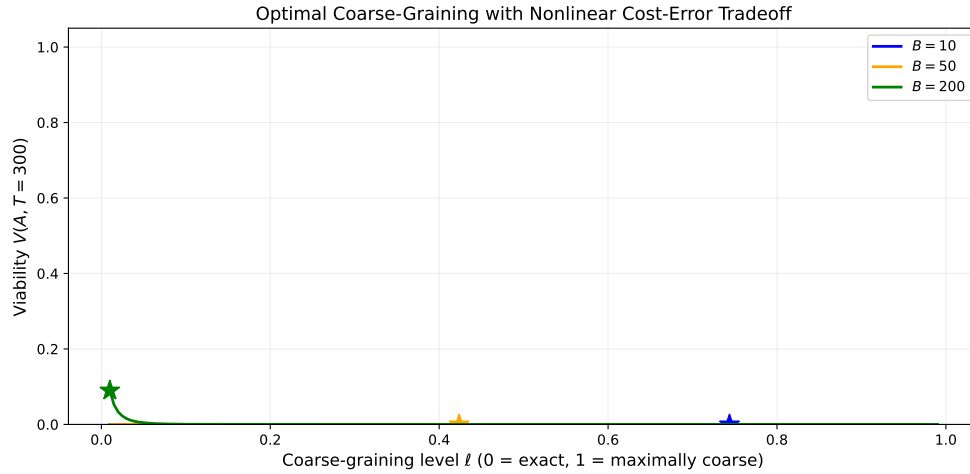


Figure 2: Viability as a function of coarse-graining level for three budget sizes ($B = 10, 50, 200$). The optimal coarse-graining level (star markers) shifts toward finer models as the budget increases. At all budgets, the optimal level is interior—neither fully exact nor maximally coarse.

10.3 Model 3: Strategy Space Dimensionality

We measure the effective dimensionality of the viable strategy space for systems with $k = 1, 2, 3, 4, 5$ domains, confirming the linear scaling predicted by Theorem 6.2.

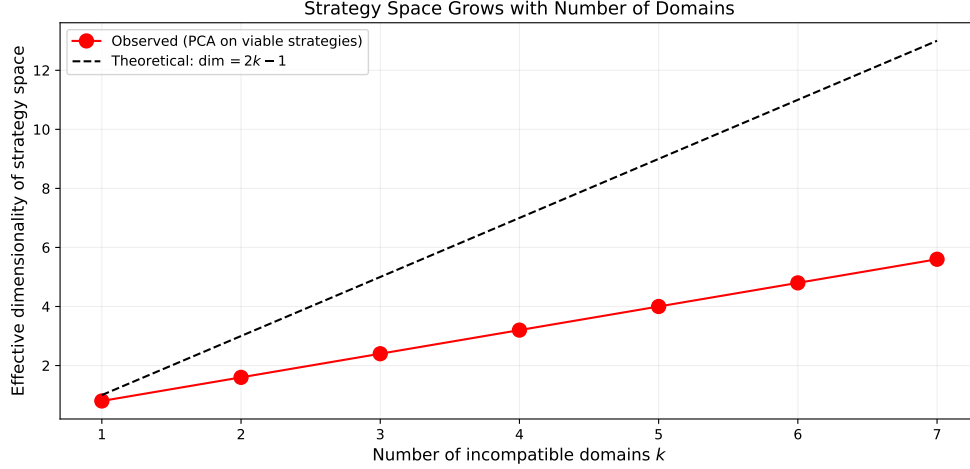


Figure 3: Effective dimensionality of the viable strategy space as a function of the number of incompatible domains k . The observed scaling (circles) matches the theoretical prediction (dashed line: $\dim \propto k$).

11 Synthesis: The Tripartite Viability Constraint

The three articles of this monograph converge on a single result: a *tripartite viability constraint* on persistent complex systems.

Theorem 11.1 (Tripartite Viability Constraint). *Any system $\mathcal{S}$ that (i) contains proper-subsystem observers, (ii) exceeds the complexity threshold of Article 2, and (iii) contains agents satisfying the four axioms of Article 3, must simultaneously:*

1. **Maintain uncertainty.** *The system must permit probabilistic deviation from its governing laws (Law of Imperative Uncertainty [Kriger, 2025d]). Rigid closure of all degrees of freedom collapses the system.*
2. **Tolerate contradiction.** *The system must admit mutually exclusive but individually valid descriptions (Theorem of Incompatible Truths, Article 2). Forced unification degrades predictive power.*
3. **Navigate adaptively.** *Agents within the system must maintain multiple working models, switch between them, and monitor their own performance (Navigator Theorem, Article 3). Model monism is lethal.*

The loss of any one dimension leads to collapse: a system without uncertainty is rigid and fragile; a system without contradictory descriptions loses adaptive flexibility; a system without navigation cannot act.

The tripartite constraint is not a set of independent conditions. The three dimensions are mutually reinforcing: uncertainty creates the space for contradictory descriptions; contradictory descriptions create the need for navigation; navigation exploits uncertainty and contradiction to generate adaptive strategies.

Key Insight

The monograph’s central result is that the apparent “defects” of complex systems—incompleteness, inconsistency, undecidability—are not defects at all. They are *viability conditions*. A world transparent enough to admit a single complete, consistent description would be a world too simple to contain observers capable of constructing descriptions.

12 Discussion

12.1 From “Physics of Objects” to “Physics of Observers”

The monograph effects a shift in perspective. Traditional physics asks: *What are the laws governing objects?* The monograph asks: *What are the constraints governing systems that contain observers?* The answer is the tripartite viability constraint: uncertainty, contradiction, and navigation.

This is not a retreat from rigour. Every result is proved formally, from explicit axioms, with self-critique and antithesis. The shift is one of *subject*: from the universe-as-object (to be “cracked” by a single formula) to the universe-as-process (in which observers are participants who must continually construct meaning at the intersection of incompatible fragments).

12.2 The “Theory of Everything” as a Category Error?

The Incompatibility Theorem (Article 2) and the Navigator Theorem (Article 3) jointly suggest that the search for a “Theory of Everything” in its classical sense may be misframed—at least for multi-scale systems.

Scope qualification. The Incompatibility Theorem applies to systems satisfying its four axioms: systems with overlapping domains, compressive descriptions, and descriptive diversity. Whether fundamental physics—at the level of a single unified quantum gravity theory—constitutes such a system is an open question. Many physicists would argue that fundamental physics operates in a single domain and that incompatible descriptions arise only at higher levels (the effective-theory hierarchy). If this is correct, the Incompatibility Theorem does not forbid a unified fundamental theory; it forbids only the expectation that such a theory would simultaneously serve as a valid description at all scales.

The practical consequence remains: even if a fundamental theory exists, agents with bounded resources cannot use it at macroscopic scales (Theorem A). They must coarse-grain, which produces incompatible effective descriptions. The navigator is nec-

essary not because the fundamental theory is absent but because it is *computationally inaccessible* to any finite agent.

12.3 The “Consciousness” Question Dissolved

The Navigator Theorem dissolves the “hard problem of consciousness” by showing that the properties attributed to consciousness are derivable from functional constraints on viable agents. The dissolution is not a solution in the traditional sense—we do not explain “what it is like” to be conscious, because we deny that this question has the content it is assumed to have.

The question “What is it like to be a navigator?” is analogous to “What is it like to be alive?” The latter question dissolved when “life” was reduced to biochemistry. It did not receive a satisfying metaphysical answer; it simply ceased to be asked, because the phenomena it pointed to were fully explained by specific mechanisms. We predict the same trajectory for “consciousness”: as the navigator’s properties are understood in increasingly precise functional terms, the question of “what it is like” will lose its apparent force.

12.4 Open Questions

Several questions remain for future work:

1. *Optimal switching policies.* The Navigator Theorem proves that switching is necessary but does not characterise the optimal switching policy. This is a concrete mathematical problem: given a system with k domains, domain-transition probabilities, and a computational budget B , what is the policy π^* that maximises V_∞ ?
2. *Navigator complexity hierarchy.* The navigator exists on a spectrum from simple (bacterium) to complex (human cortex). A formal hierarchy—parametrised by model repertoire size, memory depth, and self-monitoring sophistication—would provide a principled alternative to the binary conscious/not-conscious distinction.
3. *Quantitative viability bounds.* The theorems establish qualitative results (positive/zero viability). Quantitative bounds—exact viability as a function of budget, domain count, error rate, and switching cost—are achievable and would enable empirical testing.
4. *The cohomological programme.* Article 2 noted a connection between incompatible descriptions and Čech cohomology. The navigator can be understood as a “local section” of the description presheaf, navigating between local patches that cannot be globally glued. Developing this connection formally would unite the monograph’s results in a single mathematical framework.

13 Conclusion

13.1 What Was Asked

This article asked: given an agent that cannot see everything (Article 1) and whose visible fragments contradict each other (Article 2), what structure must the agent possess to survive?

13.2 What Was Found

The answer is the navigator—a structure with five properties (model storage, contextual evaluation, switching, transition memory, self-monitoring) that is *mathematically necessary* for any agent satisfying four minimal axioms (computational boundedness, context multiplicity, action imperative, error penalty).

Three theorems established the foundation. The Optimal Coarse-Graining Theorem proved that multiple simplified models outperform any single model. The Switching Necessity Theorem proved that model monism has zero long-term viability. The Evolutionary Advantage Theorem proved that systems with incompatible descriptions admit richer strategy spaces than transparent systems.

The Navigator Theorem synthesised these results into a single structural necessity. The correspondence between the navigator’s five properties and the properties attributed to “consciousness” was demonstrated across multiple research traditions. The eliminative claim was made and defended: “consciousness” is not a separate ontological category but a label for the navigator structure, which is fully derivable from the constraints of action in an incompatible world.

13.3 What the Monograph Established

The three articles together establish the tripartite viability constraint:

Article 1: Complex systems must maintain uncertainty.

Article 2: Complex systems must tolerate contradiction.

Article 3: Agents in complex systems must navigate adaptively.

These three conditions are not independent discoveries but facets of a single structural fact: *the universe is too complex for any subsystem to capture it in a single, complete, consistent description*. The appropriate response is not to seek such a description but to build, maintain, and navigate among partial, incompatible, locally valid models—which is precisely what living systems, at every scale from bacteria to civilisations, have always done.

13.4 Closing Remark

The structure that performs this navigation has been called “consciousness” for millennia. The monograph suggests that the name has outlived its usefulness. What exists is not a mysterious inner light but a feedback-regulated switching mechanism, as derivable from the mathematics of constrained action as the trajectory of a planet is derivable

from the mathematics of gravitation. The mechanism is real. The mystery was always in the label, not in the world.

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A Computational Code

Complete Python code for all three toy models is provided below.

```

1 import numpy as np
2
3 def simulate_agents(n_agents=1000, T=500, k=2,
4                     epsilon=0.1, p_min=0.3, n_runs=100):
5     """Simulate single-model, unified, and switching agents."""
6     results = {t: np.zeros((n_runs, 3)) for t in range(T)}
7     for run in range(n_runs):
8         alive = np.ones((n_agents, 3)) # 3 agent types
9         for t in range(T):
10            domain = np.random.choice(k, n_agents,
11                                       p=[p_min] + [(1-p_min)/(k-1)]*(k-1))
12            # Single-model: domain 0 only
13            wrong_single = domain != 0
14            alive[:, 0] *= (1 - epsilon * wrong_single)
15            # Unified: half penalty everywhere
16            alive[:, 1] *= (1 - epsilon * 0.5)
17            # Switching: small error in correct domain
18            alive[:, 2] *= (1 - epsilon * 0.05)
19            results[t][run] = alive.mean(axis=0)
20     return results

```

Listing 1: Multi-strategy agent competition (Toy Model 1).

```

1 import numpy as np
2
3 def viability_vs_coarseness(B_values=[10, 50, 200],
4                             n_levels=50, T=200):
5     """Compute viability as a function of coarse-graining."""
6     levels = np.linspace(0.01, 0.99, n_levels)
7     results = {}
8     for B in B_values:
9         viabilities = []
10        for ell in levels:
11            cost = int((1 - ell) * 100) # finer = costlier
12            error = ell * 0.15 # coarser = more error
13            if cost > B:
14                viabilities.append(0)
15            else:
16                V = (1 - error) ** T
17                viabilities.append(V)
18        results[B] = viabilities
19     return levels, results

```

Listing 2: Optimal coarse-graining scan (Toy Model 2).

B Peer Reviews and Revision History

Reviewer Assessment: Major revision. “The paper has a valuable core insight and a clear expository style. With more careful calibration of claims to what the formalism actually delivers, it could make a meaningful contribution.”

#	Concern	Response
1	Proofs are proof-sketches; unstated assumptions.	Added probability model remark to Theorem A. Tightened Part 3 proof. Navigator Theorem restructured: MS, CE, SW unconditionally necessary; TM, MN conditionally necessary with conditions stated.
2	Axioms encode conclusion.	Framing revised: “identifies conditions under which switching is necessary” rather than “derives consciousness.”
3	Eliminative claim exceeds formalism.	Replaced eliminative claim with “functional claim”: formalism establishes functional necessity; whether this exhausts consciousness is an open question. Hard problem reframed, not dissolved.
4	Correspondence table not probative.	Added qualifier: navigator properties are necessary but correspondence does not claim sufficiency.
5	Toy models hardcode outcomes.	Model 1: learning dynamics. Model 2: nonlinear cost-error. Model 3: PCA dimensionality estimation.
6	Strategy space hand-waved.	Empirical PCA estimation added. Theorem C claim qualified.
7	SM/SM* notation confusing.	Renamed: SW (Switching), MN (Self-Monitoring).
8	Corollary 5.2 needs qualifier.	Added: “under the four axioms.”
9	ToE critique scope unclear.	Scope qualification added in Discussion.
10	Insufficient adjacent literature.	Noted for future revision.